Practical 8:

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

```
~
  GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[4], time[4];
    int i,j,n=12,frames=4,hit=0,fault=0,counter=0,pos,min;

    for(i=0;i<frames;i++)
        frame[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hit++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        if(!found) {
            fault++;

            if(i < frames) {
                frame[i] = pages[i];
                counter++;
                time[i] = counter;
            }
            else {
                min = time[0];
                pos = 0;

                for(j=1;j<frames;j++) {
                    if(time[j] < min) {
                        min = time[j];
                        pos = j;
                    }
                }

                frame[pos] = pages[i];
                counter++;
                time[pos] = counter;
            }
        }
    }

    printf("Hits = %d\nFaults = %d\n", hit, fault);
    printf("Hit %% = %.2f\nFault %% = %.2f\n",(float)hit/n*100,(float)fault/n*100);

    return 0;
}
```

Output:

```
~

ASUS@LAPTOP-5DB0ADS1 MSYS ~
$ nano lru.c

ASUS@LAPTOP-5DB0ADS1 MSYS ~
$ gcc lru.c -o lru

ASUS@LAPTOP-5DB0ADS1 MSYS ~
$ ./lru
Hits = 4
Faults = 8
Hit % = 33.33
Fault % = 66.67

ASUS@LAPTOP-5DB0ADS1 MSYS ~
$ |
```